

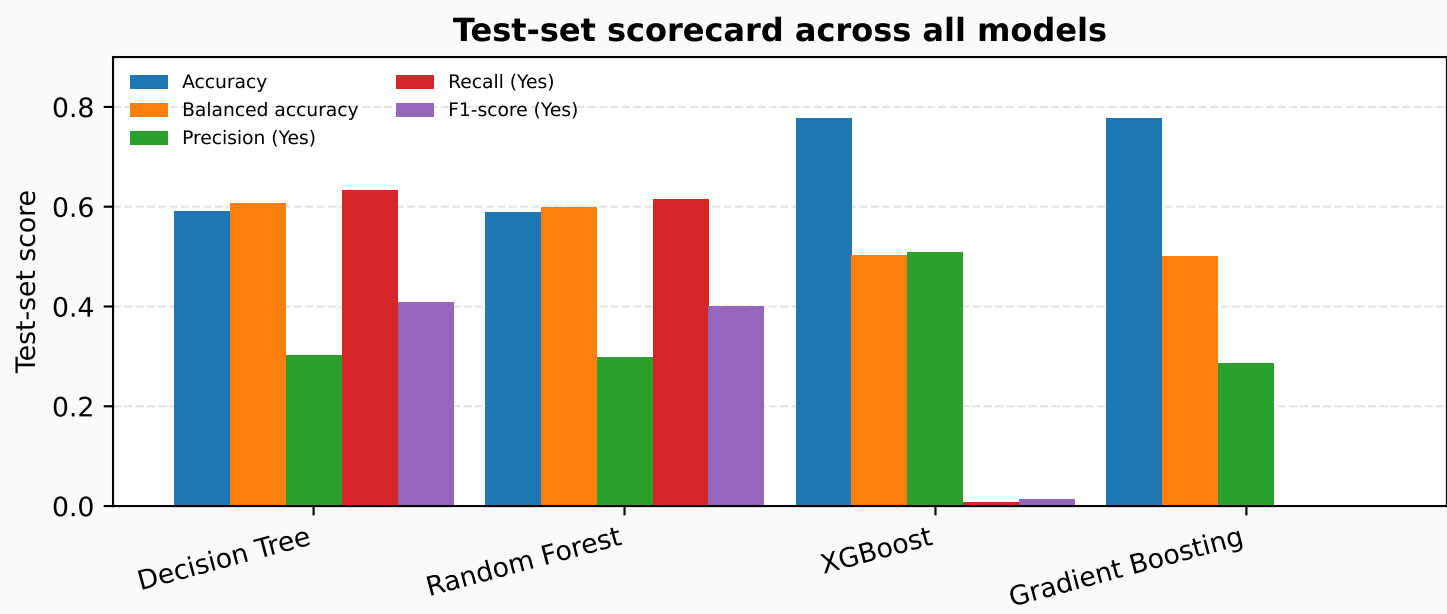
Medicine Outcome Prediction System

Professional model assessment | Business and operational view

EXECUTIVE SUMMARY

Recommended screening baseline: XGBoost

Highest accuracy and precision: XGBoost | Accuracy 0.7767 | Precision 0.5080



Decision context

XGBoost is recommended because it leads on the requested measures: accuracy (0.7767) and positive-class precision (0.5080). Its low recall (0.0072) means it should not be used as a standalone screening decision.

Evaluation design

60% training / 20% validation / 20% testing | Test set: 197,176 records | Metrics are reported for the positive outcome Yes.

Important: this is a predictive model comparison, not evidence of clinical efficacy or treatment causality. Human review and future validation remain required.

Portfolio Comparison

Testing metrics used for model selection

Model	Accuracy	Bal. Acc.	Prec. (Yes)	Recall (Yes)	F1 (Yes)	ROC-AUC	PR-AUC
Decision Tree	0.5911	0.6058	0.3017	0.6323	0.4085	0.6417	0.3260
Random Forest	0.5891	0.5983	0.2971	0.6148	0.4006	0.6326	0.3160
XGBoost	0.7767	0.5026	0.5080	0.0072	0.0142	0.6456	0.3314
Gradient Boosting	0.7766	0.5000	0.2857	0.0000	0.0001	0.6437	0.3306

Ranking by selected business objectives

Objective	Leading model	Test score
Raw accuracy	XGBoost	0.7767
Balanced accuracy	Decision Tree	0.6058
Positive recall	Decision Tree	0.6323
Positive F1-score	XGBoost	0.0142
PR-AUC	XGBoost	0.3314

Recommendation uses test-set accuracy and positive-class precision. XGBoost leads on both measures. Recall remains an important safety check because high precision alone does not identify all positive outcomes.

Decision Tree

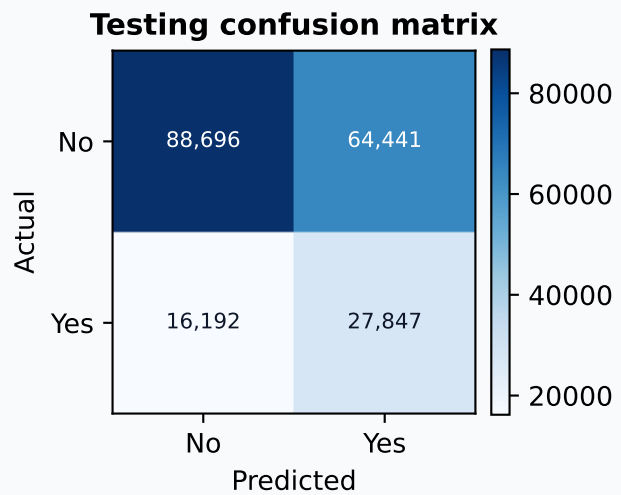
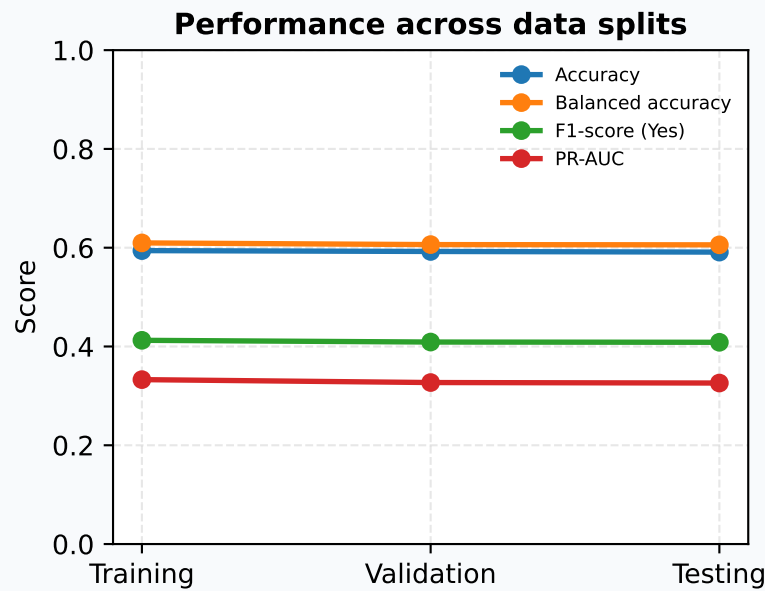
Dedicated model profile | Training, validation, and testing evidence

TEST-SET OUTCOME

Accuracy 0.5911 | Precision 0.3017 | F1 0.4085 | Recall 0.6323

Positive-class metrics refer to the Yes outcome.

Split	Accuracy	Bal. Acc.	Prec. (Yes)	Recall (Yes)	F1 (Yes)	ROC-AUC	PR-AUC
Training	0.5941	0.6097	0.3048	0.6380	0.4125	0.6501	0.3329
Validation	0.5924	0.6063	0.3024	0.6316	0.4090	0.6442	0.3270
Testing	0.5911	0.6058	0.3017	0.6323	0.4085	0.6417	0.3260



Business interpretation

Decision Tree identifies a meaningful share of positive outcomes, but produces many false positives. It is more appropriate for screening than the boosting models, with threshold tuning needed to control follow-up workload.

Test confusion matrix: TN 88,696 | FP 64,441 | FN 16,192 | TP 27,847

Random Forest

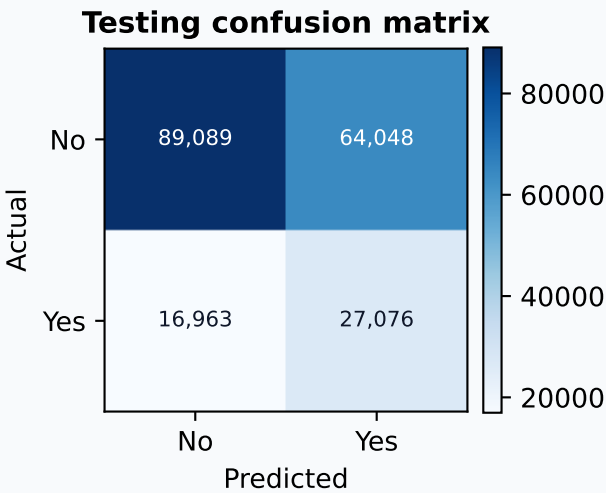
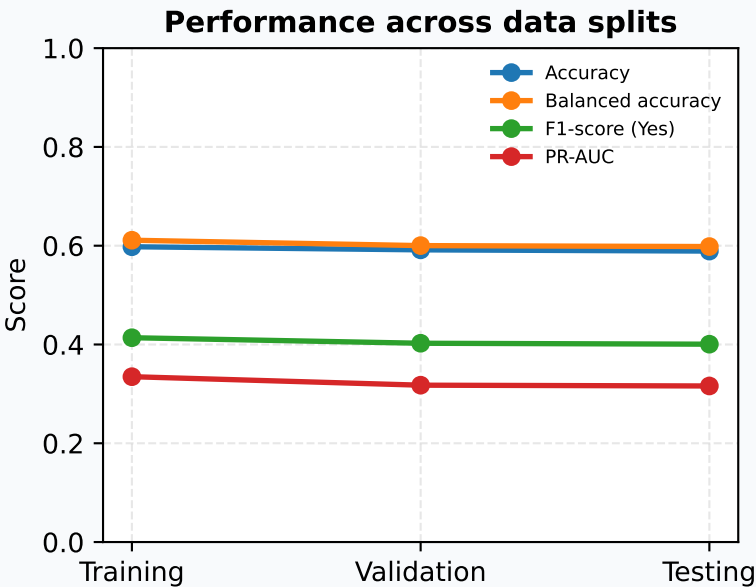
Dedicated model profile | Training, validation, and testing evidence

TEST-SET OUTCOME

Accuracy 0.5891 | Precision 0.2971 | F1 0.4006 | Recall 0.6148

Positive-class metrics refer to the Yes outcome.

Split	Accuracy	Bal. Acc.	Prec. (Yes)	Recall (Yes)	F1 (Yes)	ROC-AUC	PR-AUC
Training	0.5979	0.6111	0.3067	0.6351	0.4137	0.6513	0.3348
Validation	0.5916	0.6002	0.2989	0.6159	0.4025	0.6350	0.3176
Testing	0.5891	0.5983	0.2971	0.6148	0.4006	0.6326	0.3160



Business interpretation

Random Forest identifies a meaningful share of positive outcomes, but produces many false positives. It is more appropriate for screening than the boosting models, with threshold tuning needed to control follow-up workload.

Test confusion matrix: TN 89,089 | FP 64,048 | FN 16,963 | TP 27,076

XGBoost

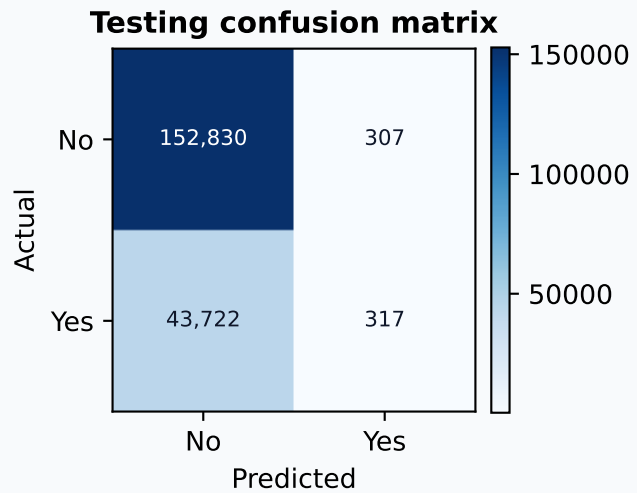
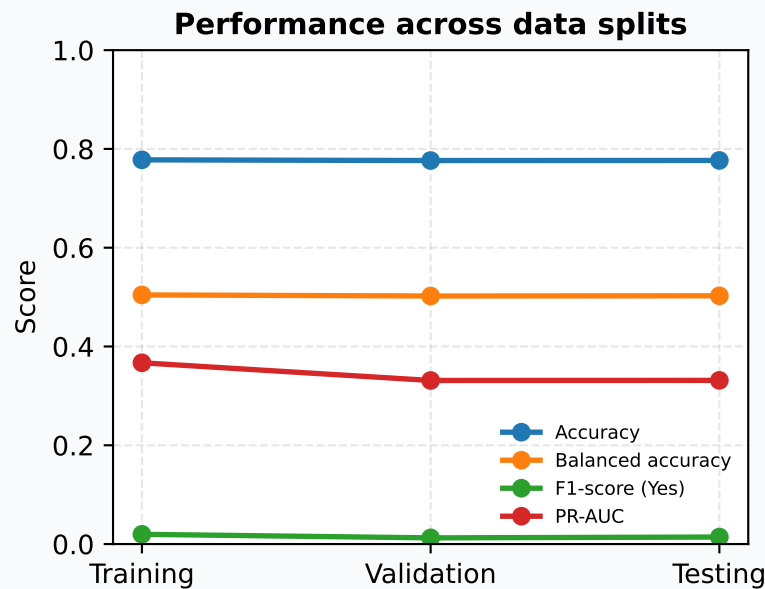
Dedicated model profile | Training, validation, and testing evidence

TEST-SET OUTCOME

Accuracy 0.7767 | Precision 0.5080 | F1 0.0142 | Recall 0.0072

Positive-class metrics refer to the Yes outcome.

Split	Accuracy	Bal. Acc.	Prec. (Yes)	Recall (Yes)	F1 (Yes)	ROC-AUC	PR-AUC
Training	0.7779	0.5044	0.6987	0.0100	0.0197	0.6720	0.3670
Validation	0.7765	0.5022	0.4786	0.0064	0.0125	0.6475	0.3312
Testing	0.7767	0.5026	0.5080	0.0072	0.0142	0.6456	0.3314



Business interpretation

XGBoost achieves high accuracy because it predicts the majority No class. Its near-zero positive recall means it is not suitable for a screening workflow at the current decision threshold.

Test confusion matrix: TN 152,830 | FP 307 | FN 43,722 | TP 317

Gradient Boosting

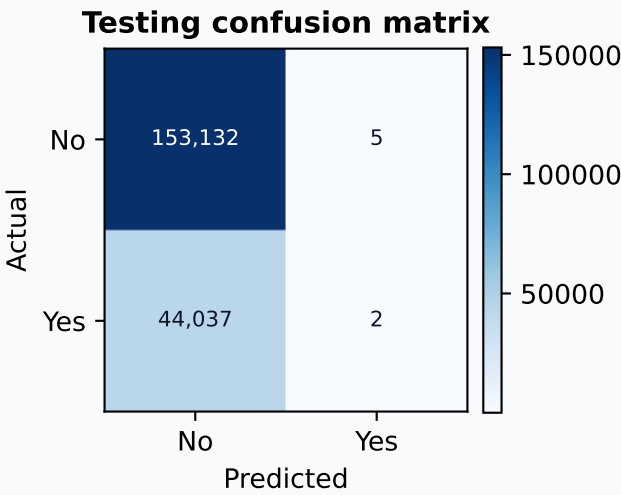
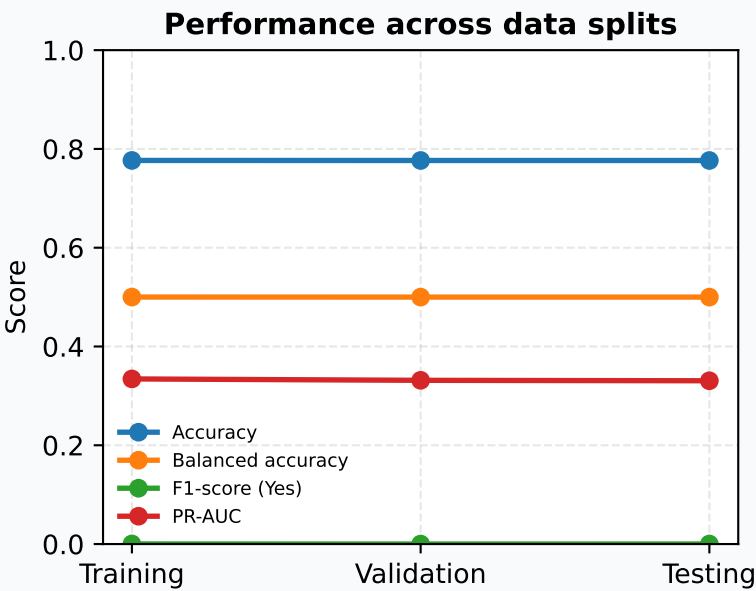
Dedicated model profile | Training, validation, and testing evidence

TEST-SET OUTCOME

Accuracy 0.7766 | Precision 0.2857 | F1 0.0001 | Recall 0.0000

Positive-class metrics refer to the Yes outcome.

Split	Accuracy	Bal. Acc.	Prec. (Yes)	Recall (Yes)	F1 (Yes)	ROC-AUC	PR-AUC
Training	0.7767	0.5001	0.9333	0.0001	0.0002	0.6470	0.3343
Validation	0.7766	0.5000	0.2500	0.0000	0.0000	0.6463	0.3315
Testing	0.7766	0.5000	0.2857	0.0000	0.0001	0.6437	0.3306



Business interpretation

Gradient Boosting achieves high accuracy because it predicts the majority No class. Its near-zero positive recall means it is not suitable for a screening workflow at the current decision threshold.

Test confusion matrix: TN 153,132 | FP 5 | FN 44,037 | TP 2

Final Recommendation and Controls

Decision summary for stakeholders

Recommended model

Use XGBoost for controlled validation because it leads on test-set accuracy (0.7767) and positive-class precision (0.5080).

Safety conclusion

Although XGBoost leads on accuracy and precision, its positive recall is only 0.0072. Use human review, threshold analysis, and recall monitoring before any screening deployment.

Before deployment

Tune the decision threshold against the cost of false negatives and false positives.
Validate calibration, subgroup performance, temporal stability, and performance on new data.

Governance

Keep human review in the loop. Log predictions, review errors, monitor drift, and define rollback and periodic revalidation procedures.

Limitation

This report evaluates predictive performance only. It does not establish clinical efficacy, causal treatment benefit, fairness, privacy compliance, or readiness for patient-facing decisions.

Source: four model treatment-outcome metric files in the output directory.